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WHAT IS ACTUALLY UNDER 3-6-9

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[1] THE SAME MACHINE, ONE FLOOR DOWN — repeating decimals

The period of 1/p is exactly the order of 10 in the group mod p.
Digital roots and repeating decimals are the SAME fact.

p	period of 1/p	p-1	full reptend?
7	6	6	YES
11	2	10	no
13	6	12	no
17	16	16	YES
19	18	18	YES
23	22	22	YES
29	28	28	YES
31	15	30	no
37	3	36	no
41	5	40	no
43	21	42	no
47	46	46	YES

When the period is maximal (p-1) the prime is called FULL REPTEND,
and its decimal expansion is a CYCLIC NUMBER. Watch 1/7:

```
142857 x 1 =  142857
142857 x 2 =  285714
142857 x 3 =  428571
142857 x 4 =  571428
142857 x 5 =  714285
142857 x 6 =  857142
142857 x 7 =  999999  <-- all nines
```

Every multiple from 1 to 6 is the SAME SIX DIGITS, rotated.
The seventh is 999999. This is the genuine version of the magic
people think they see in 3-6-9 — and it is completely explained:
10 is a primitive root mod 7, so its powers sweep every residue
once before returning, and each sweep is a rotation of the same cycle.

[2] WHICH MODULI EVEN HAVE A PRIMITIVE ROOT? (Gauss)

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have one : [2, 3, 4, 5, 6, 7, 9, 10, 11, 13, 14, 17, 18, 19, 22, 23, 25, 26, 27, 29, 31, 34, 37, 38] have none: [8, 12, 15, 16, 20, 21, 24, 28, 30, 32, 33, 35, 36, 39]

Gauss proved the pattern exactly: only 1, 2, 4, p^k and $2p^k$ for odd primes p. Our $9 = 3^2$ makes the list — which is the reason doubling sweeps the whole group and the 3-6-9 split appears. Change 9 for 8, 12, 15 or 16 and the structure collapses.

[3] THE UNSOLVED PROBLEM HIDING IN THE DOUBLING MAP

Ask the obvious next question: for how many primes p is 2 a primitive root — i.e. how often does doubling sweep everything? Artin conjectured in 1927 that the density is

$$A = \prod_p \left(1 - \frac{1}{p(p-1)}\right) = 0.3739558136$$

NOBODY HAS PROVED IT. Hooley proved it in 1967 assuming the Generalised Riemann Hypothesis. Unconditionally it is still open. We can only measure it:

primes up to	count	2 is prim.root	density	vs A
1,000	167	67	0.4012	+0.0272
10,000	1,228	470	0.3827	+0.0088
50,000	5,132	1,923	0.3747	+0.0008
100,000	9,591	3,603	0.3757	+0.0017
200,000	17,983	6,707	0.3730	-0.0010
300,000	25,996	9,701	0.3732	-0.0008

It converges on Artin's constant and nobody can explain why. That is the honest floor beneath 3-6-9: not a hidden key, but a hundred-year-old open problem in the same machinery.

[4] THE UNIFICATION — this is where our whole session meets

Every structure we have touched is CHARACTERS OF A GROUP:

group		character	we call them	leads to
circle	R/Z	$e^{2 \pi i t}$	sin, cos	Fourier series
cube	$(Z/2)^n$	prod x_i	parity functions	the census, Huang
units	$(Z/n)^*$	chi(a)	Dirichlet characters	L-functions, RH

Same construction three times. The first gave you sin and cos. The second gave us Parseval and the 65,536-function census. The third gives Dirichlet characters, then L-functions, then the Riemann Hypothesis. One idea, three floors, and the bottom floor is the deepest open problem in mathematics.

characters of $(Z/9)^*$, the group that generates 3-6-9:

```
    group [1, 2, 4, 5, 7, 8], generator 2, order 6
    chi_0: exponents of w=e^(2 pi i/6) ->  0  0  0  0  0  0
    chi_1: exponents of w=e^(2 pi i/6) ->  0  1  2  5  4  3
    chi_2: exponents of w=e^(2 pi i/6) ->  0  2  4  4  2  0
    (the j=0 row is the trivial character; the rest encode the
     multiplicative structure the digital-root pattern rides on)

[5] WHERE IT REALLY IS A KEY — the same arithmetic, working
--------------------------------------------------------------------
  primes p,q      1,000,003 , 1,000,033
  modulus n       1,000,036,000,099
  phi(n)          1,000,034,000,064
  public  e       65537
  private d       983,264,276,609

  encrypt 369  -> 801,004,867,329
  decrypt      -> 369   recovered

  That is Euler's theorem: a^phi(n) = 1 mod n. The identical fact
  that makes digital roots work makes this work. Its security rests
  on factoring being hard — a complexity assumption, which is where
  this whole conversation started.

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  3-6-9 is a shadow cast by base ten.
  The thing casting it runs from repeating decimals through an
  unsolved conjecture to the Riemann Hypothesis and RSA.
  The shadow is not the key. The lamp is.
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